

41. Let $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{b} = \hat{i} + \hat{j} - \hat{k}$. If $\vec{c}$ is a vector such that $\vec{a} \cdot \vec{c} = 11$, $\vec{b} \cdot (\vec{a} \times \vec{c}) = 27$ and $\vec{b} \cdot \vec{c} = -\sqrt{3}|\vec{b}|$, then $|\vec{a} \times \vec{c}|^2$ is equal to _____.

माना $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$ तथा $\vec{b} = \hat{i} + \hat{j} - \hat{k}$ हैं। यदि एक सदिश $\vec{c}$ के लिए $\vec{a} \cdot \vec{c} = 11$, $\vec{b} \cdot (\vec{a} \times \vec{c}) = 27$ तथा $\vec{b} \cdot \vec{c} = -\sqrt{3}|\vec{b}|$ है, तो $|\vec{a} \times \vec{c}|^2$ बराबर है _____

Ans. 285

Sol. $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b} = \hat{i} + \hat{j} - \hat{k}$

$$\vec{b} \cdot (\vec{a} \times \vec{c}) = 27, \vec{a} \cdot \vec{b} = 0$$

$$\vec{b} \times (\vec{a} \times \vec{c}) = -3\vec{a}$$

Let θ be angle between $\vec{b}$, $\vec{a} \times \vec{c}$

$$\text{Then } |\vec{b}| \cdot |\vec{a} \times \vec{c}| \sin \theta = 3\sqrt{14}$$

$$|\vec{b}| \cdot |\vec{a} \times \vec{c}| \cos \theta = 27$$

$$\Rightarrow \sin \theta = \frac{\sqrt{14}}{\sqrt{95}}$$

$$\therefore |\vec{b}| \cdot |\vec{a} \times \vec{c}| = 3\sqrt{95}$$

$$\Rightarrow |\vec{a} \times \vec{c}| = \sqrt{3} \times \sqrt{95}$$

42. If $\vec{a} = 2\hat{i} + \hat{j} + 2\hat{k}$, then the value of $|\hat{i} \times (\vec{a} \times \hat{i})|^2 + |\hat{j} \times (\vec{a} \times \hat{j})|^2 + |\hat{k} \times (\vec{a} \times \hat{k})|^2$ is equal to _____.

यदि $\vec{a} = 2\hat{i} + \hat{j} + 2\hat{k}$ है, तो $|\hat{i} \times (\vec{a} \times \hat{i})|^2 + |\hat{j} \times (\vec{a} \times \hat{j})|^2 + |\hat{k} \times (\vec{a} \times \hat{k})|^2$ का मान है _____ ।

Ans. 18

Sol. Let $\vec{a} = x\hat{i} + y\hat{j} + z\hat{k}$

$$\hat{i} \times (\vec{a} \times \hat{i}) = (\hat{i} \cdot \hat{i}) \vec{a} - (\vec{a} \cdot \hat{i}) \hat{i}$$

$$= x\hat{i} + y\hat{j} + z\hat{k} - x\hat{i}$$

$$= y\hat{j} + z\hat{k} \quad \text{.....(i)}$$

$$\text{similarly } \hat{j} \times (\vec{a} \times \hat{j}) = x\hat{i} + z\hat{k} \quad \text{.....(ii)}$$

$$\text{and } \hat{k} \times (\vec{a} \times \hat{k}) = x\hat{i} + y\hat{j} \quad \text{.....(iii)}$$

$$|\hat{y}\hat{j} + z\hat{k}|^2 + |x\hat{i} + z\hat{k}|^2 + |x\hat{i} + y\hat{j}|^2$$

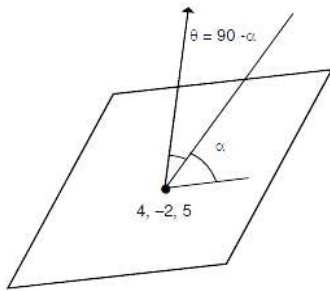
$$2(x^2 + y^2 + z^2) = 2|\vec{a}|^2 = 18$$

43. Let θ be the angle between the planes $P_1 : \vec{r} \cdot (\hat{i} + \hat{j} + 2\hat{k}) = 9$ and $P_2 : \vec{r} \cdot (2\hat{i} - \hat{j} + \hat{k}) = 15$. Let L be the line that meets P_2 at the point $(4, -2, 5)$ and makes an angle θ with the normal of P_2 . If α is the angle between L and P_2 , then $(\tan^2 \theta)(\cot^2 \alpha)$ is equal to _____.

माना समतलों $P_1 : \vec{r} \cdot (\hat{i} + \hat{j} + 2\hat{k}) = 9$ तथा $P_2 : \vec{r} \cdot (2\hat{i} - \hat{j} + \hat{k}) = 15$ के बीच का कोण θ है। माना रेखा L समतल P_2 को बिंदु $(4, -2, 5)$ पर मिलती है तथा P_2 के अभिलंब से कोण θ बनाती है। यदि L तथा P_2 के बीच कोण α है तो $(\tan^2 \theta)(\cot^2 \alpha)$ बराबर है।

Ans. 9

Sol.



$$\cos \theta = \frac{(\hat{i} + \hat{j} + 2\hat{k}) \cdot (2\hat{i} - \hat{j} + \hat{k})}{6} = \frac{2 - 1 + 2}{6} = \frac{1}{2}$$

$$\theta = \frac{\pi}{3} \quad \alpha = \frac{\pi}{6}$$

$$\therefore (\tan^2 \theta)(\cot^2 \alpha) = (3)(3) = 9$$

44. Let P be an arbitrary point having sum of the squares of the distances from the planes $x + y + z = 0$, $lx - nz = 0$ and $x - 2y + z = 0$, equal to 9. If the locus of the point P is $x^2 + y^2 + z^2 = 9$, then the value of $l - n$ is equal to _____.

माना P एक स्वेच्छ बिन्दु जिसकी समतलों $x + y + z = 0$, $lx - nz = 0$ तथा $x - 2y + z = 0$ से दूरियों के वर्गों का योगफल 9 है। यदि बिन्दु P का बिन्दुपथ $x^2 + y^2 + z^2 = 9$ है, तो $l - n$ का मान बराबर है _____।

Ans. 0

Sol. Let P (α, β, γ)

$$d_1 + d_2 + d_3 = 9$$

$$\left(\frac{\alpha + \beta + \gamma}{\sqrt{3}}\right)^2 + \left(\frac{\ell\alpha - n\gamma}{\sqrt{\ell^2 + n^2}}\right)^2 + \left(\frac{\alpha - 2\beta + \gamma}{\sqrt{1 + 4 + 1}}\right)^2 = 9$$

Now Locus is

$$\left(\frac{x + y + z}{\sqrt{3}}\right)^2 + \left(\frac{\ell x - nz}{\sqrt{\ell^2 + n^2}}\right)^2 + \left(\frac{x - 2y + z}{\sqrt{6}}\right)^2 = 9$$

$$\left(\frac{1}{2} + \frac{\ell^2}{\ell^2 + n^2}\right) + y^2 \left(\frac{1}{3} + \frac{4}{6}\right) + z^2 \left(\frac{1}{2} + \frac{\ell^2}{\ell^2 + n^2}\right) + \alpha z x \left(\frac{1}{2} - \frac{\ell n}{\ell^2 + n^2}\right) = 9$$

but Locus is $x^2 + y^2 + z^2 = 9$

By comparison

$$\frac{1}{2} + \frac{\ell^2}{\ell^2 + n^2} = 1 \quad \dots\dots\dots(1)$$

$$\frac{1}{2} - \frac{\ell n}{\ell^2 + n^2} = 0 \quad \dots\dots\dots(2)$$

by (1) & (2)

$$\frac{\ell^2}{\ell^2 + n^2} = \frac{1}{2}$$

$$\frac{\ell n}{\ell^2 + n^2} = \frac{1}{2}$$

we get $\ell = n$ hence $\ell - n = 0$

45. Let the line $\ell : x = \frac{1-y}{-2} = \frac{z-3}{\lambda}, \lambda \in \mathbb{R}$ meet the plane P : $x + 2y + 3z = 4$ at the point (α, β, γ). If the angle

between the line l and the plane P is $\cos^{-1}\left(\sqrt{\frac{5}{14}}\right)$, then $\alpha + 2\beta + 6\gamma$ is equal to _____.

माना रेखा $\ell : x = \frac{1-y}{-2} = \frac{z-3}{\lambda}, \lambda \in \mathbb{R}$ समतल P : $x + 2y + 3z = 4$ को बिंदु (α, β, γ) पर मिलती है। यदि रेखा l तथा समतल

P के बीच का कोण $\cos^{-1}\left(\sqrt{\frac{5}{14}}\right)$ है, तो $\alpha + 2\beta + 6\gamma$ बराबर है _____

Ans. 11

Sol. $\ell : x = \frac{y-1}{2} = \frac{z-3}{\lambda}, \lambda \in \mathbb{R}$

Dr's of line $\ell(1, 2, \lambda)$

Dr's of normal vector of plane P : $x + 2y + 3z = 4$
are (1, 2, 3)

Now, angle between line ℓ and plane P is given by

$$\sin \theta = \left| \frac{1+4+3\lambda}{\sqrt{5+\lambda^2} \cdot \sqrt{14}} \right| = \frac{3}{\sqrt{14}} \left(\text{given } \cos \theta = \sqrt{\frac{5}{14}} \right)$$

$$\Rightarrow \lambda = \frac{2}{3}$$

Let variable point of line ℓ is $\left(t, 2t+1, \frac{2}{3}t+3\right)$

line of plane P.

$$\Rightarrow t = -1$$

$$\Rightarrow \left(-1, -1, \frac{7}{3}\right) \equiv (\alpha, \beta, \gamma)$$

$$\Rightarrow \alpha + 2\beta + 6\gamma = 11$$

46. Let $\vec{a} = 3\hat{i} + \hat{j} - \hat{k}$ and $\vec{c} = 2\hat{i} - 3\hat{j} + 3\hat{k}$. If $\vec{b}$ is a vector such that $\vec{a} = \vec{b} \times \vec{c}$ and $|\vec{b}|^2 = 50$, then $|72 - |\vec{b} + \vec{c}|^2|$ is equal to _____.

माना $\vec{a} = 3\hat{i} + \hat{j} - \hat{k}$ तथा $\vec{c} = 2\hat{i} - 3\hat{j} + 3\hat{k}$ हैं। यदि एक सदिश $\vec{b}$ इस प्रकार है कि $\vec{a} = \vec{b} \times \vec{c}$ तथा $|\vec{b}|^2 = 50$ हैं, तो $|72 - |\vec{b} + \vec{c}|^2|$ बराबर है _____

Ans. 66

Sol. $|\vec{a}| = |\vec{b} \times \vec{c}|$
 $= \sqrt{11} = |\vec{b}| \cdot \sqrt{22} \cdot \sin \theta$
 $\therefore \sin \theta = \frac{1}{10}$

$$\text{or } \cos \theta = \frac{\sqrt{99}}{10}$$

$$\begin{aligned} |72 - |\vec{b} + \vec{c}|^2| &= \left| 72 - \left(50 + 22 + 2 \times 5\sqrt{2} \cdot \sqrt{22} \frac{\sqrt{99}}{10} \right) \right| \\ &= \left| 72 - \left(72 + \frac{2 \times 5 \times 11 \times 3}{10} \right) \right| \\ &= |66| = 66 \end{aligned}$$

47. Let the co-ordinates of one vertex of ΔABC be $A(0, 2, \alpha)$ and the other two vertices lie on the line $\frac{x+\alpha}{5} = \frac{y-1}{2} = \frac{z+4}{3}$. For $\alpha \in \mathbb{Z}$, if the area of ΔABC is 21 sq. units and the line segment BC has length $2\sqrt{21}$ units, then α^2 is equal to _____.

माना ΔABC के एक शीर्ष के निर्देशांक $A(0, 2, \alpha)$ है तथा अन्य दो शीर्ष रेखा $\frac{x+\alpha}{5} = \frac{y-1}{2} = \frac{z+4}{3}$ पर स्थित है। $\alpha \in \mathbb{Z}$ के लिए, यदि ΔABC का क्षेत्रफल 21 वर्ग इकाई है एवं रेखा खण्ड BC की लम्बाई $2\sqrt{21}$ इकाई है, तब α^2 बराबर है।

Ans. 9

Sol. A. $(0, 2, \alpha)$

$(-\alpha, 1, -4) \quad B \quad C \quad (5i + 2j + 3k)$

$$\left| \frac{1}{2} \cdot 2\sqrt{21} \cdot \begin{vmatrix} i & j & k \\ \alpha & 1 & \alpha+4 \\ 5 & 2 & 3 \end{vmatrix} \right| \frac{1}{\sqrt{25+4+9}} = 21\sqrt{21}$$

$$\begin{aligned} \sqrt{(2\alpha+5)^2 + (2\alpha+20)^2 + (2\alpha-5)^2} &= \sqrt{21} \sqrt{38} \\ \Rightarrow 12\alpha^2 + 80\alpha + 450 &= 798 \\ \Rightarrow 12\alpha^2 + 80\alpha - 348 &= 0 \\ \Rightarrow \alpha = 3 \Rightarrow \alpha^2 &= 9 \end{aligned}$$

48. The point of intersection C of the plane $8x + y + 2z = 0$ and the line joining the points $A(-3, -6, 1)$ and $B(2, 4, -3)$ divides the line segment AB internally in the ratio $k : 1$. If a, b, c ($|a|, |b|, |c|$ are coprime) are the direction ratios of the perpendicular from the point C on the line $\frac{1-x}{1} = \frac{y+4}{2} = \frac{z+2}{3}$, then $|a + b + c|$ is equal to _____.

समतल $8x + y + 2z = 0$ तथा बिंदुओं $A(-3, -6, 1)$ और $B(2, 4, -3)$ को मिलाने वाली रेखा का प्रतिच्छेदन बिंदु C रेखाखंड AB को $k : 1$ के अनुपात में अंतः विभाजित करता है। यदि बिंदु C से रेखा $\frac{1-x}{1} = \frac{y+4}{2} = \frac{z+2}{3}$ पर अभिलंब के दिक् अनुपात a, b, c ($|a|, |b|, |c|$ असहभाज्य हैं) हैं, तो $|a + b + c|$ बराबर है।

Ans. 10

Sol. $C\left(\frac{2k-3}{k+1}, \frac{4k-6}{k+1}, \frac{-3k+1}{k+1}\right)$

C lies on $8x + y + 2z = 0$

$$\therefore 8(2k-3) + (4k-6) + 2(-3k+1) = 0$$

$$16k - 24 + 4k - 6 - 6k + 2 = 0$$

$$14k = 28 \Rightarrow k = 2$$

$$\therefore C \text{ is } \left(\frac{1}{3}, \frac{2}{3}, \frac{-5}{3}\right)$$

Any point on line $\frac{x-1}{-1} = \frac{y+4}{2} = \frac{z+2}{2}$ is $D(1-\lambda, -4+2\lambda, -2+3\lambda)$

$$\text{Drs of } CD = 1-\lambda - \frac{1}{3}, -4+2\lambda - \frac{2}{3}, -2+3\lambda + \frac{5}{3}$$

$$= \frac{2-3\lambda}{3}, \frac{-14+6\lambda}{3}, \frac{-1+9\lambda}{3}$$

$$CD \perp \text{ to line : } -1(2-3\lambda) + 2(6\lambda-14) + 3(9\lambda-1) = 0$$

$$\Rightarrow -2 + 3\lambda + 12\lambda - 28 + 27\lambda - 3 = 0$$

$$42\lambda = 33 \Rightarrow \lambda = \frac{33}{42} = \frac{11}{14}$$

$$\therefore \text{drs of } CD = \frac{2-\frac{33}{14}}{3}, \frac{-14+\frac{66}{14}}{3}, \frac{-1+\frac{99}{14}}{3}$$

$$= -5, -130, 85$$

$$= -1, -26, 17$$

$$\therefore (-1 - 26 + 17) = 10$$

49. Let the equation of the plane passing through the line $x - 2y - z - 5 = 0 = x + y + 3z - 5$ and parallel to the line $x + y + 2z - 7 = 0 = 2x + 3y + z - 2$ be $ax + by + cz = 65$. Then the distance of the point (a, b, c) from the plane $2x + 2y - z + 16 = 0$ is _____.

माना रेखा $x - 2y - z - 5 = 0 = x + y + 3z - 5$ से होकर जाने वाले तथा रेखा $x + y + 2z - 7 = 0 = 2x + 3y + z - 2$ के समांतर समतल का समीकरण $ax + by + cz = 65$ है। तो बिंदु (a, b, c) की समतल $2x + 2y - z + 16 = 0$ से दूरी है।

Ans. 9

Sol. since plane passing through the line

$$x - 2y - z - 5 = 0 = x + y + 3z - 5$$

So equation of plane is

$$(x - 2y - z - 5) + \lambda (x + y + 3z - 5) = 0$$

$$(1 + \lambda)x + (\lambda - 2)y + (3\lambda - 1)z - 5 - 5\lambda = 0 \quad (1)$$

And this plane parallel to the line $x + y + 2z - 7 = 0 = 2x + 3y + z - 2$ with direction ratio $(-5, 3, 1)$

Now

$$-5(1 + \lambda) + (\lambda - 2) \cdot 3 + (3\lambda - 1) \cdot 1 = 0 \text{ [since plane \& line are parallel]}$$

$$\Rightarrow \lambda = 12$$

Hence equation of plane is $13x + 10y + 35z = 65$

Given equation of plane is $ax + by + cz = 65$

so, $(a, b, c) = (13, 10, 35)$

Now we have to find the distance of $(13, 10, 35)$ from $2x + 2y - z + 16 = 0$

$$d = \frac{|26 + 20 - 35 + 16|}{\sqrt{4 + 4 + 1}} = 9$$